

What Is Planetary Defense?

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Planetary defense is the practical work of finding asteroids and comets near Earth, calculating where they are going, communicating risk clearly, and preparing response options before a real threat becomes urgent.

The simple definition

Planetary defense protects Earth from natural impact hazards. It combines sky surveys, observatories, orbital calculations, spacecraft engineering, public communication, and emergency planning.

The goal is not fear. The goal is time: time to confirm the science, time to warn correctly, and time to act if action is ever needed.

Close approach does not mean impact

Most near-Earth object passes are routine scientific events. Risk depends on size, distance, speed, uncertainty, and probability - not just whether an object appears in a close-approach list.

How planetary defense works

- ? 1. ****Discover**** - survey telescopes and observers detect a moving object against the background stars.
- ? 2. ****Follow up**** - additional observations reduce uncertainty and help confirm that the object is real.
- ? 3. ****Refine the orbit**** - scientists calculate the object's path and compare future close approaches with Earth.
- ? 4. ****Assess risk**** - distance, size, velocity, and probability determine whether the event is routine or needs attention.
- ? 5. ****Communicate clearly**** - public updates should be sourced, calm, and separated from official emergency instructions.
- ? 6. ****Prepare options**** - if evidence warrants it, options can include more tracking, spacecraft missions, deflection studies, or civil preparedness.

What TESS supports

- ? Public education and mission dashboards.
- ? Observatory readiness and practical sky-monitoring workflows.
- ? Responsible communication that points back to authoritative sources.

What this page is not

- ? It is not an official government alert.
- ? It is not an impact prediction.
- ? It does not replace NASA/JPL, IAWN, or local emergency-management guidance.